



HELIOS

How the System Works

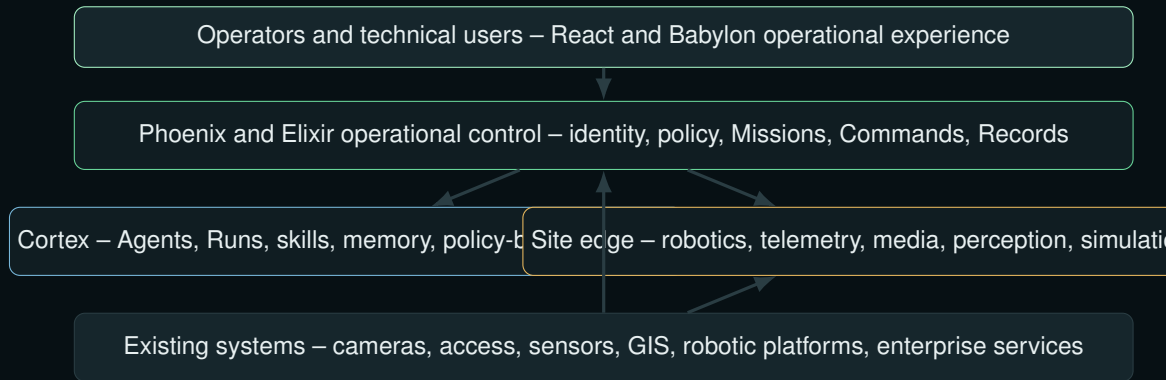
Operational world, Cortex, Missions, robotic assets, and resilient Site cells

CONTROLLED TECHNICAL OVERVIEW

Architecture and engineering evaluation edition

Version 0.1 – August 2026

System at a glance



Recommended baseline

Site-local operational continuity with central enterprise coordination. Existing systems remain authoritative for the facts and controls they own.

One authority path. Deliberate technical boundaries.

Operational control plane owns

Organization and Site identity; Physical and Simulation Operations; policy; Mission Definition, Revision, Proposal, Run, and Tasks; resource reservation; Command admission; evidence relationships; audit; public interfaces.

Site and edge own

Vendor protocols; target execution; platform-local rejection and failsafes; high-rate telemetry; media transport; perception; simulation and measured native workloads.

Cortex owns

Agent identity and replaceable Runs; provider supervision; bounded tools; assessment; recommendations; memory retrieval; policy-permitted intent; structured Agent requests.

Paths that do not exist

The browser, scheduler, or Cortex cannot directly operate a platform. Simulation identity cannot become physical execution. Adapters cannot create Mission authority. Client state cannot become operational truth.

Constitution

Every human, schedule, Cortex, policy, Agent, or tool request that may produce an effect enters the same BEAM-owned admission path.

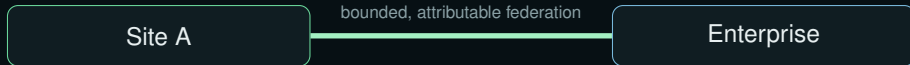
Resilient Site cells with enterprise coordination

SITE-LOCAL HELIOS

- ▶ Local operator access and durable state.
- ▶ Current operational world and approved Missions.
- ▶ Site Cortex and bounded local policy.
- ▶ Robotic adapters, telemetry, media, and perception.
- ▶ Evidence buffering and local observability.
- ▶ Bounded operation during enterprise disconnection.

ENTERPRISE HELIOS

- ▶ Multi-Site posture and Attention.
- ▶ Organization policy and administration.
- ▶ Central Cortex and longer-horizon analysis.
- ▶ Cross-Site requests and resource coordination.
- ▶ Federated Records, memory, and learning.
- ▶ Approved package, skill, model, and policy distribution.



Disconnection never creates new authority. Reconnect reconciles existing facts and never silently duplicates effects.

The operational world is authoritative context

World Revision

Source digests; geometry; coordinate frames; units; transforms; calibration; uncertainty; semantic objects; zones; routes; POIs; sensors; deployments; limitations; publication and commissioning state.

Presentation package

Babylon or another approved renderer consumes a derived package. Renderer state may optimize presentation but cannot become canonical world truth.

Source geometry and surveys



Frames, calibration, uncertainty



Semantic objects and policy zones



Routes, sensors, assets, readiness



Missions, evidence, operational state

When the system is in a state of readiness, a rendered scene is n

Mission and Command: intent to observed effect



Mission structure

Reusable Definition; immutable Revision; exact Proposal; target preparation; Execution Authorization Snapshot; one admitted Run; independently progressing Task Runs; attributable completion and Record.

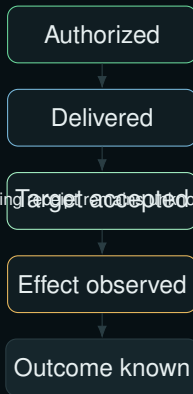
Command truth is orthogonal

Authority, delivery, target acceptance, observed effect, and Mission consequence remain separate. One dimension cannot silently advance another.

Unknown result

If a Command may have reached a target but the response is lost, Helios reconciles the existing attempt. It does not blindly issue another effect.

Why sent is not the same as happened



Each transition needs its own attributable fact. A missing target results in a stale state.

No blind replay

A lost response does not create permission to issue another physical effect.

One current controller

Operator, Cortex Run, Site process, and edge execution all obey current generation or lease authority.

Record what was known

The Record preserves intent, authority, delivery, acceptance, effect, uncertainty, intervention, and outcome.

Cortex: hybrid intelligence with fenced Runs

Durable Agent

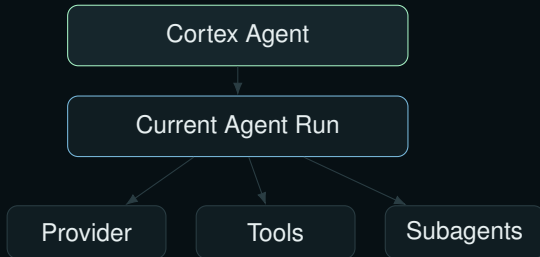
Accountable identity, role, scope, profile revision, authority ceiling, skill and tool policy, memory scope, and human supervisor.

Replaceable Agent Run

Generation, model or provider, approved context, skills, tools, budget, audience, consequence class, checkpoint, lease, start and stop state. A newer generation fences an older Run.

Phase 3 autonomy

Current policy + current evidence + exact Site and Mission class + current readiness + eligible resources + preserved reserve coverage → ordinary Mission and Command admission → operator supervision and intervention.



Media, integration, identity, and trust

Media and perception

Camera or sensor → Site ingest → selected live viewing and source-authoritative recording. Perception produces structured detections, Tracks, observations, confidence, and provenance.

Cortex input

Structured semantic output, approved stills or clips, evidence references, source health, spatial context, uncertainty, and policy. Continuous raw video remains in the media path.

Integration patterns

Attributable events; snapshot plus delta; bounded source query; media reference; native command boundary; evidence export. Realtime messages signal change and clients reload authoritative projections after gaps.

Identity and trust

Human, service, device, and Agent identities remain distinct. Service communication should be mutually authenticated and encrypted. Browsers and model providers never receive reusable source or platform secrets.

Evidence rule

Sensor reading, detection, Track, observation, entity association, Alert, Incident, evidence, assessment, recommendation, decision, Command, and outcome remain distinct attributable facts.

Degraded operation, delivery, and status

Failure behavior

- ▶ Source outage: unavailable with freshness.
- ▶ Media outage: no negative inference.
- ▶ Cortex outage: native operator controls remain.
- ▶ Driver outage: resource degrades or becomes unavailable.
- ▶ Enterprise outage: bounded Site-local operation continues.
- ▶ Database or authority uncertainty: writes fail closed.
- ▶ Unknown effect: reconcile, never blind replay.

Visible delivery path

Foundation: identity, event, mode, and no-bypass controls.

Phase 1: complete operator-directed robotic Mission and Exercise.

Phase 2: Cortex assessment, recommendations, memory, and learning candidates.

Phase 3: policy-bounded autonomous observation.

Phase 4: structured multi-Agent and cross-Site resource coordination.

Current technical posture

CURRENT Operator, spatial, execution, media, event, topology, Cortex, and authority foundations **IN**
DEVELOPMENT Mission, Command, readiness, edge integration, and operational Cortex **PLANNED**